A MAP OF INVERTIBLE SHEAVES INDUCES A NORM MAP ON THEIR NORMS, AND VANISHING OF THE MAP IS DETECTED AT A POINT EXACTLY WHEN THE NORM MAP VANISHES THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-norm-map-invertible is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3695-3715 and proof span 3717-3735 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Reread under the 2026-10-14 prefilter rule: this candidate is 856 bytes over 19 proof lines and the prefilter matched the string 'Some details omitted.' in the last line of the proof. On rereading, that sentence refers to the cocycle verification of the definition just given, while the proof itself supplies the trivialization, the local equations, the definition of the norm map and the reduction of the final equivalence to condition (2) of the definition of a norm map; the candidate was therefore accepted, in the same way as the previously counted card S13-morph-relative-normalization-dvr, and the remaining omission is recorded here rather than counted as a gap in the statement.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-norm-map-invertible resolves in the project's own tag list to canonical tag 0BCZ (line 14297 of the index, full label divisors-lemma-norm-map-invertible). Since reference/public-items.jsonl holds no row whose primary source_file is divisors.tex, the same-chapter public tag id set is empty and the intersection with the resolved tag id is empty, so the candidate passes.

Editorial note. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source divisors.tex –label lemma-norm-map-invertible –start 3695 –end 3715 –proof-start 3717 –proof-end 3735 –local-id 34088, giving RESULT=OK with the label found inside the statement span, disjoint statement and proof spans, exactly one proof environment in the gap between them, distinct statement and proof text, no public interval row for this chapter and occupancy FREE for divisors.tex 3695 3735. Spans were recomputed live by locating the label in the pinned chapter file,

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and the located text matched the pool byte counts (statement 654 characters, proof 749 characters after whitespace removal).

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Invertible sheaves become trivial after a finite pullback along the source of a finite morphism, cited from the same chapter. Cited at proof lines 3718-3721. Produces the open covering of the target over whose preimages the two invertible sheaves are simultaneously trivial, which is what makes the local definition of the norm map possible.
- Construction of the norm of an invertible sheaf out of a norm of degree d, cited from the same chapter. Cited at statement line 3709 and proof lines 3727-3730. Supplies the target objects of the induced map and the cocycle description whose compatibility has to be verified.
- Condition (2) in the definition of a norm map of degree d, cited from the same chapter. Cited at proof lines 3732-3733. Converts the pointwise statement about the local equation f_j into the stated equivalence between vanishing of the map and vanishing of its norm.

Source reference. Original lemma 0BUT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14886–14886. Same-chapter reference used in the retained author text at source line(s) 3673, 3721. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BCY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14296–14296. Same-chapter reference used in the retained author text at source line(s) 3709, 3730. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes. Assume there exists a norm of degree d for π. For any $\mathcal{O}_X$-linear map $\varphi : \mathcal{L} \to \mathcal{L}'$ of invertible $\mathcal{O}_X$-modules there is an $\mathcal{O}_Y$-linear map*

$$Norm_\pi(\varphi) : Norm_\pi(\mathcal{L}) \longrightarrow Norm_\pi(\mathcal{L}')$$

with $Norm_\pi(\mathcal{L})$, $Norm_\pi(\mathcal{L}')$ as in Lemma 0BCY. Moreover, for $y \in Y$ the following are equivalent

(1) *φ is zero at a point of $x \in X$ with $\pi(x) = y$, and*
(2) *$Norm_\pi(\varphi)$ is zero at y.*

Proof. We choose an open covering $Y = \bigcup V_j$ such that $\mathcal{L}$ and $\mathcal{L}'$ are trivial over the opens $\pi^{-1}V_j$. This is possible by Lemma 0BUT. Choose generating sections s_j and s'_j of $\mathcal{L}$ and $\mathcal{L}'$ over the opens $\pi^{-1}V_j$. Then $\varphi(s_j) = f_j s'_j$ for some $f_j \in \mathcal{O}_X(\pi^{-1}V_j)$. Define $\mathrm{Norm}_\pi(\varphi)$ to be multiplication by $\mathrm{Norm}_\pi(f_j)$ on V_j. A simple calculation involving the cocycles used to construct $\mathrm{Norm}_\pi(\mathcal{L})$, $\mathrm{Norm}_\pi(\mathcal{L}')$ in the proof of Lemma 0BCY shows that this defines a map as stated in the lemma. The final statement follows from condition (2) in the definition of a norm map of degree d. Some details omitted. □